

CV – MAX MELCHING

PERSONAL INFORMATION

Email maxm.melching@gmail.com
GitHub [MaxMelching](https://github.com/MaxMelching)
Website <https://maxmelching.github.io>

EDUCATION

09/2025 – present PhD (Physics) California Institute of Technology
10/2022 – 05/2025 M.Sc. (Physics) Leibniz University Hannover
FINAL GRADE: 1,0 (with honors) ✦ Equivalent GPA: 4.0
MINOR: Mathematics
10/2018 – 10/2022 B.Sc. (Physics) Leibniz University Hannover
FINAL GRADE: 1,0 (with honors) ✦ Equivalent GPA: 4.0
MINOR: Computer Science

WORK EXPERIENCE

10/2024 – 08/2025 Student Assistant Max Planck Institute for Gravitational Physics
Various tasks related to the Max Planck Leadnet, such as website development and maintenance, and organizational duties.
REFERENCE: Frank Ohme ✦ frank.ohme@aei.mpg.de
04/2023 – 07/2024 Student Research Assistant Institute for Quantum Optics, Hannover
Data analysis and software development for the Cold Atom Lab experiment onboard the ISS. Part of that was the development of a PYTHON package that can be used to read, process and evaluate experimental data.
REFERENCES: Naceur Gaaloul ✦ gaaloul@iqo.uni-hannover.de, Gabriel Müller ✦ g.mueller@iqo.uni-hannover.de
10/2021 – 02/2022 Student Assistant Leibniz University Hannover
I was one of the tutors for an introductory physics lecture for first-semester undergraduate students. Responsibilities included grading exercise sheets and exams, as well as teaching tutorials to part of the class.
REFERENCE: Tammo Block ✦ block@maphy.uni-hannover.de

PUBLICATIONS & THESES

2026 Paper (soon to be published) [arXiv:2604.21859](https://arxiv.org/abs/2604.21859)
TITLE: Mitigating Systematic Errors in Parameter Estimation of Binary Black Hole Mergers in O1-O3 LIGO-Virgo Data
DESCRIPTION: Applies the method developed in the "Accounting for the known unknowns" paper to binary black hole mergers that showed systematic differences in their source parameters when analyzed with different

waveform models. We specifically look at the events from the first three LIGO-Virgo-KAGRA observing runs (O1-O3).

AUTHOR LIST: Sumit Kumar, Frank Ohme, Max Melching, Harsh Narola, Tom Dooney, Chris Van Den Broeck

2025 *Paper (published in PRD)* [arXiv:2502.17400](#)

TITLE: Accounting for the known unknowns: A parametric framework to incorporate systematic waveform errors in gravitational-wave parameter estimation

DESCRIPTION: Proposes a new way to account for systematic errors in gravitational waveform models and thereby also mitigate the associated biases.

AUTHOR LIST: Sumit Kumar, Frank Ohme, Max Melching

11/2023 – 05/2025 *Master Thesis* Max Planck Institute for Gravitational Physics

TITLE: Systematic Errors in Gravitational Waveform Models

DESCRIPTION: Theoretical development and extension, as well as implementation, of tools in the Fisher matrix framework that aims at quantifying waveform systematics.

SUPERVISORS: Frank Ohme ✦ frank.ohme@aei.mpg.de, Krishnendu NV ✦ krishnendu.nv@icts.res.in

DOI: [10.15488/19415](#)

03/2022 – 10/2022 *Bachelor Thesis* Max Planck Institute for Gravitational Physics

TITLE: Systematic Differences in the Source Properties of the Third Gravitational-Wave Catalog

DESCRIPTION: Analysis of the posterior distributions of the third catalog of gravitational wave events, with an emphasis on waveform model systematics.

SUPERVISOR: Frank Ohme ✦ frank.ohme@aei.mpg.de

DOI: [10.15488/19414](#)

HONORS & AWARDS

2023/2024 Deutschlandstipendium ✦ Leibniz University Hannover (monthly stipend for two semesters)

2022/2023 Deutschlandstipendium ✦ Leibniz University Hannover (monthly stipend for two semesters)

2020 Niedersachsenstipendium ✦ Leibniz University Hannover (one-time payment stipend)

OTHER INFORMATION

COMPUTER SKILLS

ADVANCED: PYTHON, L^AT_EX, git (includes GitLab, GitHub), Jupyter

INTERMEDIATE: Linux, Mathematica, C, Microsoft Windows

BASIC: slurm, condor, MATLAB, Microsoft Office

PRESENTATIONS

- 09/2024 Poster “Using Correlations for Good – Systematic Errors Using Alignment” at the LIGO-Virgo-KAGRA Meeting in Barcelona (LIGO DCC: [G2401544](#))
- 03/2024 Oral Presentation “Systematic Differences in the Source Properties Of Gravitational Wave Signals” at the DPG Spring Meeting of the German physical society, section Gravitation and Relativity
- 04/2024 Oral Presentation “Systematic Errors in Gravitational Waveform Models” at the DPG Spring Meeting of the German physical society, section Gravitation and Relativity

PERSONAL

LANGUAGES: German (Native) ♦ English (Full Working Proficiency)

May 10, 2026